

# Transcriptomic and Network Analysis of Alzheimer's Disease Hippocampal Gene Expression

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## ABSTRACT

Alzheimer's disease (AD) affects the hippocampus early, but gene expression changes in advanced disease are hard to detect in small post-mortem studies. We analyzed public microarray data (GSE1297; Affymetrix HG-U133A; 22,283 probes) from 9 control and 7 severe AD hippocampal samples to identify genes, biological processes and protein interactions that differ between the groups. Expression values were log2-transformed and compared using limma with Benjamini–Hochberg false discovery rate (FDR) correction. Although 3,482 probes had a nominal  $P < 0.05$ , no probe reached an adjusted  $P < 0.05$  (minimum adjusted  $P = 0.117$ ), so the 150 probes with the smallest nominal p-values (143 unique genes) were used as an exploratory set for Gene Ontology (GO) and KEGG enrichment with Enrichr and for a protein–protein interaction (PPI) network in STRING v12. No GO term (1,273 tested) or KEGG pathway (175 tested) was significant after correction, with lowest adjusted P-values of 0.095 and 0.42. The leading GO terms related to growth regulation, miRNA-related processes and sex-hormone biology, while synaptic terms ranked lower. The STRING network contained 137 nodes and 81 edges (PPI enrichment  $P = 0.060$ ), and 64 nodes (47%) were unconnected. The most connected genes were NRXN1 (8 connections), CSF2 (7), GATA3 (7), GFAP (6) and CD68 (6). With only 16 samples, no individual gene was significant. The most connected genes include synaptic (NRXN1) and glial (GFAP, CD68) genes, but these findings are exploratory and need confirmation in a larger independent dataset.

**Keywords:** Alzheimer's disease; hippocampus; microarray; limma; protein–protein interaction network; GSE1297

## 1. Introduction

Alzheimer's disease (AD) is the most common form of dementia, contributing to an estimated 60–70% of cases among the 57 million people who had dementia worldwide in 2021, with nearly 10 million new cases each year (WHO, 2026). It is a progressive neurodegenerative disorder characterized by cognitive decline, synaptic loss (Selkoe, 2002) and neuroinflammation (Heneka et al., 2015). Hallmark pathological features such as amyloid plaques and neurofibrillary tangles (NFTs) are well documented, and NFT pathology reaches limbic regions including the hippocampus at early stages of disease (Braak & Braak, 1991). However, the molecular mechanisms driving disease progression, particularly at early, incipient stages, remain incompletely understood. Because early-stage AD markers substantially overlap with normal aging, identifying reliable transcriptional signatures of disease progression has proven difficult (Blalock et al., 2004).

Transcriptomic analysis of post-mortem brain tissue offers one approach to this problem. Blalock et al. (2004) addressed this challenge directly, profiling hippocampal gene expression across 31 subjects (9 cognitively normal controls and 22 AD patients spanning incipient to severe stages) and correlating expression with two measures of AD severity: Mini-Mental State Examination (MMSE) scores and post-mortem neurofibrillary tangle counts. Their

analysis identified a large transcriptional response associated with AD progression, implicating growth-related and tumor-suppressor pathways. This dataset (GEO accession GSE1297) has since been reanalyzed by multiple independent groups using different computational approaches (Liang et al., 2018; Gómez Ravetti et al., 2010; Sun et al., 2019). Notably, a later reanalysis using information-theoretic and combinatorial optimization methods identified a 1,372-probe gene signature strongly associated with AD severity, highlighting alterations in calcium, insulin, phosphatidylinositol and Wnt signaling, with the most severity-correlated probes linked to synaptic function, neurofilament assembly, and neuronal plasticity (Gómez Ravetti et al., 2010). Separately, weighted gene co-expression network analysis (WGCNA) applied to the same dataset identified AD-associated co-expression modules and candidate hub genes, including metallothionein, Notch2, MSX1, ADD3 and RAB31 (Liang et al., 2018) and GRIK1 (Sun et al., 2019), underscoring the value of network-based approaches beyond single-gene statistics for extracting biological signal from this dataset.

Building on this body of work, the present study applied differential expression analysis, functional enrichment analysis, and protein–protein interaction network analysis to the same GSE1297 dataset, focusing on the comparison between control and severe AD samples. Earlier reanalyses concentrated on correlations with severity measures, feature selection, or co-expression modules; here we used a transparent, reproducible pipeline that explicitly accounts for the limited statistical power of a 16-sample comparison. The aims were to identify differentially expressed genes between groups, to assess whether the top-ranked genes were enriched for specific biological processes or pathways, and to construct a protein–protein interaction network to identify the most highly connected proteins within the gene set.

## **2. Methods**

### **2.1 Dataset**

Gene expression data were obtained from the Gene Expression Omnibus (GEO; accession GSE1297) using the GEOquery package (Davis & Meltzer, 2007) in R. GSE1297 is a publicly available microarray dataset profiling human hippocampal tissue using the Affymetrix Human Genome U133A (HG-U133A) platform (GPL96) (Blalock et al., 2004). It contains samples from 31 donors: healthy controls ( $n = 9$ ) and patients with Alzheimer's disease (AD) at incipient, moderate, and severe stages. For this analysis, only the Control ( $n = 9$ ) and Severe AD ( $n = 7$ ) groups were selected, giving 16 samples. This comparison was chosen to maximize the biological contrast between unaffected and advanced AD tissue while retaining sufficient samples for statistical analysis.

### **2.2 Data preprocessing**

The scale of the expression values was inspected before differential expression analysis. The values were log2-transformed to reduce right-skewness and stabilize variance across the intensity range, making them appropriate for the linear modeling framework used by limma. The resulting expression matrix contained 22,283 probes, all used for genome-wide differential expression analysis.

### **2.3 Differential expression analysis**

Differential expression between the Control and Severe AD groups was assessed using the limma package (Ritchie et al., 2015; version 3.68.4) in R (version 4.6.1). Limma fits a linear model independently for each probe and applies empirical Bayes moderation to stabilize variance estimates across genes, an approach particularly appropriate for gene expression datasets with relatively small sample sizes. A design matrix representing the two experimental groups was used to model the Control versus Severe AD comparison. For each probe, the analysis produced a nominal p-value, a moderated test statistic, and a multiple-testing-adjusted p-value. Probes were ranked by nominal p-value, and significance was assessed using the Benjamini–Hochberg false discovery rate (FDR; Benjamini & Hochberg, 1995), with adjusted  $P < 0.05$  as the threshold. Because no probe met this threshold, the 150 probes with the smallest nominal p-values were retained as an exploratory ranked gene set for downstream analyses. Inclusion in this set was not considered equivalent to statistical significance.

## 2.4 Probe annotation and gene selection

The selected Affymetrix probe identifiers were mapped to official gene symbols using the hgu133a.db Bioconductor annotation package (version 3.13.0). Probes without a gene symbol were excluded, and where several probes mapped to the same gene symbol, duplicate gene entries were removed to obtain a unique gene set. This yielded 143 uniquely annotated genes from the 150 top-ranked probes, which were used as input for both enrichment and network analysis.

## 2.5 Functional enrichment analysis

Enrichment analysis was performed with Enrichr (Kuleshov et al., 2016) against the GO Biological Process 2026 and KEGG 2026 libraries. Enrichment was assessed from the overlap between the submitted genes and genes annotated to each term, using the p-values and Benjamini–Hochberg-adjusted p-values reported by Enrichr. Because the input set came from nominal p-value ranking and not from genes meeting  $FDR < 0.05$ , the analysis was treated as exploratory, and the highest-ranked terms were examined to identify biological themes represented in the set.

## 2.6 Protein–protein interaction network analysis

The 143-gene set was submitted to STRING v12.0 (Szklarczyk et al., 2023), restricted to *Homo sapiens*, with a minimum interaction confidence of 0.400. The network was used to examine functional relationships among the encoded proteins. Network-level enrichment was assessed with STRING's statistical framework to identify biological processes and molecular functions among the interacting proteins. Node degree, the number of interaction edges per protein, was used to identify the most connected nodes (candidate "hub" genes).

## 2.7 Publication Enrichment Analysis

Publication enrichment was performed in STRING on the genes in the PPI network, against the STRING default whole-genome background, to identify PubMed publications that mention an unusually large share of these genes. Results were interpreted as literature overlap, not biological validation.

## 2.8 Analytical approach

Differential expression was evaluated using FDR-adjusted p-values. Because no gene met this threshold, the top-150 list was used only as an exploratory input for the enrichment and

network analyses, and results were interpreted as patterns across the gene set, not as evidence for individual genes.

## 2.9 Software and resources

Data retrieval and differential expression analysis were performed using R version 4.6.1. Functional enrichment and protein–protein interaction analyses were performed using Enrichr and STRING, respectively. The principal software and databases used were:

| Tool/Database | Version        | Purpose                            |
|---------------|----------------|------------------------------------|
| R             | 4.6.1          | Statistical analysis               |
| GEOquery      | 2.80.0         | GEO dataset retrieval              |
| limma         | 3.68.4         | Differential expression analysis   |
| hgu133a.db    | 3.13.0         | Affymetrix probe annotation        |
| Enrichr       | 2026 libraries | GO and KEGG enrichment             |
| STRING        | 12.0           | PPI network and network enrichment |
| GEO           | GSE1297        | Source expression dataset          |

## 3. Results

### 3.1 Differential Gene Expression Analysis

Gene expression profiles were compared between 9 control and 7 severe AD hippocampal samples using limma across 22,283 probes. A total of 3,482 probes had a nominal  $P < 0.05$ , compared with about 1,114 expected by chance if no probe were differentially expressed. However, no probe met the significance threshold after Benjamini–Hochberg correction (adjusted  $P < 0.05$ ); the smallest adjusted  $P$ -value was 0.117 (probe 215306\_at; log<sub>2</sub> fold change = 1.834; Figure 1, Table 1). No individual gene was therefore identified as differentially expressed between control and severe AD hippocampus. The 150 probes with the smallest nominal  $P$ -values were annotated, yielding 143 unique genes (Section 2.4), and the downstream enrichment and network results are interpreted as exploratory.

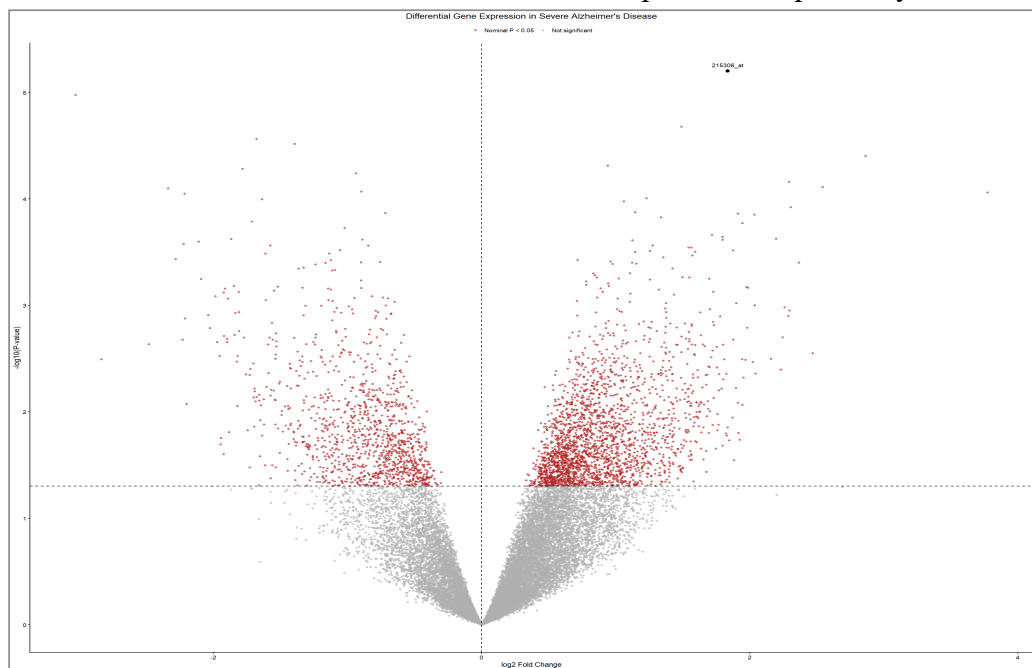


Figure 1. Volcano plot showing differential expression between control and severe Alzheimer's disease hippocampal samples. Each point represents an analyzed probe. The x-axis represents log2 fold change and the y-axis represents  $-\log_{10}(\text{P-value})$ . Probes with nominal  $P < 0.05$  are highlighted. The most statistically significant probe, 215306\_at, is labelled. No probe reached an FDR-adjusted  $P < 0.05$ .

Table 1. Top 20 ranked probes from the differential gene expression analysis.

| Rank | Probe ID    | logFC  | Average Expression | t-statistic | P-value               | Adjusted P-value (FDR) |
|------|-------------|--------|--------------------|-------------|-----------------------|------------------------|
| 1    | 215306_at   | 1.834  | 8.218              | 6.372       | $6.26 \times 10^{-6}$ | 0.1172                 |
| 2    | 215161_at   | -3.030 | 7.515              | -6.109      | $1.05 \times 10^{-5}$ | 0.1172                 |
| 3    | 217675_at   | 1.49   | 2.96               | 5.766       | $2.09 \times 10^{-5}$ | 0.1302                 |
| 4    | 211496_s_at | -1.680 | 5.399              | -5.638      | $2.72 \times 10^{-5}$ | 0.1302                 |
| 5    | 219073_s_at | -1.398 | 7.664              | -5.583      | $3.04 \times 10^{-5}$ | 0.1302                 |
| 6    | 215917_at   | 2.863  | 3.969              | 5.457       | $3.94 \times 10^{-5}$ | 0.1302                 |
| 7    | 219746_at   | 0.939  | 8.687              | 5.358       | $4.84 \times 10^{-5}$ | 0.1302                 |
| 8    | 206232_s_at | -1.786 | 6.429              | -5.324      | $5.20 \times 10^{-5}$ | 0.1302                 |
| 9    | 213666_at   | -0.938 | 9.364              | -5.278      | $5.73 \times 10^{-5}$ | 0.1302                 |
| 10   | 206278_at   | 2.292  | 5.645              | 5.189       | $6.89 \times 10^{-5}$ | 0.1302                 |
| 11   | 215320_at   | 2.543  | 2.942              | 5.136       | $7.71 \times 10^{-5}$ | 0.1302                 |
| 12   | 204055_s_at | -2.342 | 5.709              | -5.123      | $7.93 \times 10^{-5}$ | 0.1302                 |
| 13   | 218801_at   | -0.899 | 8.858              | -5.090      | $8.50 \times 10^{-5}$ | 0.1302                 |
| 14   | 202018_s_at | 3.772  | 7.058              | 5.08        | $8.69 \times 10^{-5}$ | 0.1302                 |
| 15   | 205344_at   | -2.217 | 7.508              | -5.067      | $8.93 \times 10^{-5}$ | 0.1302                 |
| 16   | 210229_s_at | 1.229  | 5.031              | 5.022       | $9.80 \times 10^{-5}$ | 0.1302                 |

|    |           |        |       |        |                       |        |
|----|-----------|--------|-------|--------|-----------------------|--------|
| 17 | 219042_at | -1.639 | 7.643 | -5.010 | $1.01 \times 10^{-4}$ | 0.1302 |
| 18 | 213929_at | 1.06   | 7.582 | 4.989  | $1.05 \times 10^{-4}$ | 0.1302 |
| 19 | 211606_at | 2.305  | 5.567 | 4.929  | $1.19 \times 10^{-4}$ | 0.1361 |
| 20 | 216809_at | 1.144  | 7.194 | 4.878  | $1.33 \times 10^{-4}$ | 0.1361 |

### 3.2 Gene Ontology Enrichment Analysis

The 143-gene set was tested against 1,273 GO Biological Process terms. The top-ranked term was negative regulation of cell growth (overlap 6/124;  $P = 2.69 \times 10^{-4}$ ; adjusted  $P = 0.0947$ ), followed by positive regulation of miRNA transcription (4/48;  $P = 3.82 \times 10^{-4}$ ) and male gonad development (4/49;  $P = 4.14 \times 10^{-4}$ ). The other leading terms were development of primary male sexual characteristics, negative regulation of dendrite development (2/5), positive regulation of miRNA metabolic process, androgen metabolic process, positive regulation of neurotransmitter uptake (2/6) and macrophage differentiation (Table 2). The ten leading terms shared the same adjusted P-value of 0.0947. Terms related to neuronal function ranked lower: positive regulation of synaptic transmission (26th; 4/88; adjusted  $P = 0.18$ ), regulation of dendrite development (29th), nervous system development (33rd; 10/531), positive regulation of synaptic transmission, glutamatergic (70th) and glutamate receptor signaling pathway (120th). No GO term reached significance after multiple-testing correction (lowest adjusted  $P = 0.0947$ ).

Table 2. Top GO Biological Process enrichment terms

| Rank | GO Biological Process                              | Overlap | P-value               | Adjusted P-value (FDR) | Genes                                        |
|------|----------------------------------------------------|---------|-----------------------|------------------------|----------------------------------------------|
| 1    | Negative Regulation of Cell Growth                 | 6/124   | $2.69 \times 10^{-4}$ | 0.0947                 | SMAD3; BCL11A; WT1; CAPRIN2; INHBA; KIAA0319 |
| 2    | Positive Regulation of miRNA Transcription         | 4/48    | $3.82 \times 10^{-4}$ | 0.0947                 | SMAD3; WT1; GATA3; GATA2                     |
| 3    | Male Gonad Development                             | 4/49    | $4.14 \times 10^{-4}$ | 0.0947                 | SRD5A1; WT1; GATA3; INHBA                    |
| 4    | Development of Primary Male Sexual Characteristics | 4/50    | $4.47 \times 10^{-4}$ | 0.0947                 | SRD5A1; WT1; GATA3; INHBA                    |
| 5    | Negative Regulation of Dendrite Development        | 2/5     | $5.01 \times 10^{-4}$ | 0.0947                 | BCL11A; KIAA0319                             |

|    |                                                                 |       |                            |        |                                                                        |
|----|-----------------------------------------------------------------|-------|----------------------------|--------|------------------------------------------------------------------------|
| 6  | Positive Regulation of<br>miRNA Metabolic<br>Process            | 4/54  | 6.00 ×<br>10 <sup>-4</sup> | 0.0947 | SMAD3; WT1;<br>GATA3;<br>GATA2                                         |
| 7  | Androgen Metabolic<br>Process                                   | 3/24  | 6.49 ×<br>10 <sup>-4</sup> | 0.0947 | SRD5A1;<br>AKR1C4;<br>CYP3A4                                           |
| 8  | Positive Regulation of<br>Neurotransmitter<br>Uptake            | 2/6   | 7.47 ×<br>10 <sup>-4</sup> | 0.0947 | ITGB1; DRD4                                                            |
| 9  | Macrophage<br>Differentiation                                   | 3/26  | 8.25 ×<br>10 <sup>-4</sup> | 0.0947 | CSF2; CSF1;<br>GATA3                                                   |
| 10 | Positive Regulation of<br>Macromolecule<br>Biosynthetic Process | 9/344 | 8.41 ×<br>10 <sup>-4</sup> | 0.0947 | CSF2; SMAD3;<br>CSF1; WT1;<br>NRXN1;<br>INHBA;<br>GATA2;<br>PLCB1; TTN |

### 3.3 KEGG Pathway Enrichment Analysis

The 143-gene set was tested against 175 KEGG pathways. The pathway with the strongest nominal enrichment was transcriptional misregulation in cancer (6/189; nominal P = 0.0024; adjusted P = 0.42), followed by Virion – Lassa virus and SFTS virus (2/17), which comprised only PCDH11X and PCDH11Y, steroid hormone biosynthesis (3/63), bacterial invasion of epithelial cells (3/78) and cell adhesion molecules (4/155; including NRXN1) (Table 3, Figure 2). Neurodegeneration-related pathways ranked lower: glutamatergic synapse was 43rd (P ≈ 0.19) and pathways of neurodegeneration – multiple diseases was 55th (P ≈ 0.25). No pathway reached significance after correction (lowest adjusted P = 0.42).

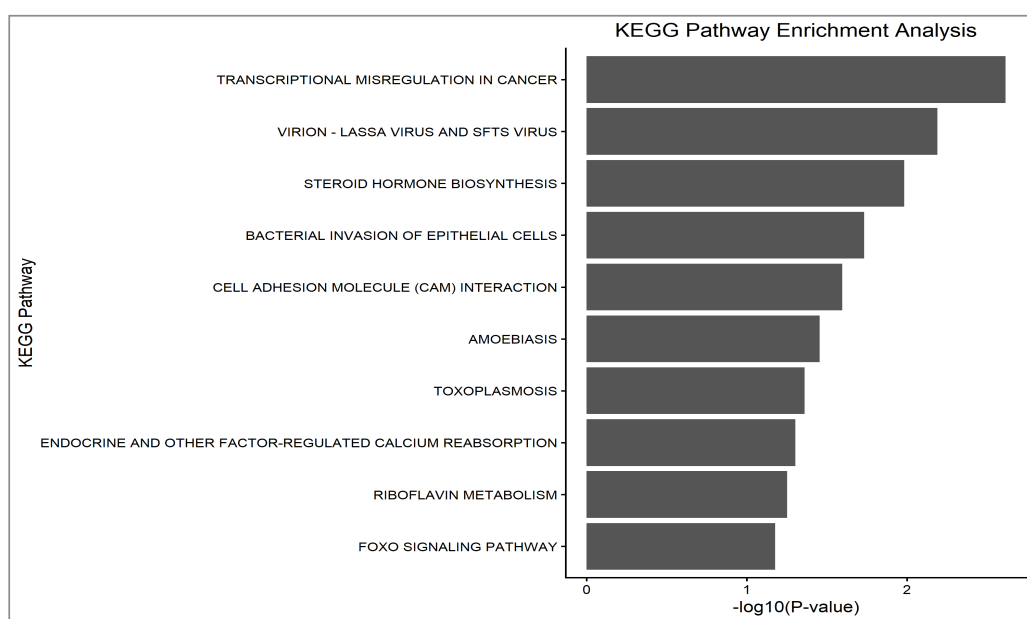


Figure 2. KEGG pathway enrichment analysis of the analyzed gene set. Top 10 enriched KEGG pathways are shown according to nominal P-value, represented as  $-\log_{10}(P\text{-value})$ .

Table 3. Top enriched KEGG pathways with overlap, P-value, adjusted P-value, and associated genes.

| Rank | KEGG Pathway                                              | Overlap | P-value | Adjusted P-value | Genes                              |
|------|-----------------------------------------------------------|---------|---------|------------------|------------------------------------|
| 1    | Transcriptional Misregulation in Cancer                   | 6/189   | 0.00241 | 0.42203          | ARNT2; CSF2; WT1; PAX3; ETV1; LDB2 |
| 2    | Virion – Lassa Virus and SFTS Virus                       | 2/17    | 0.00643 | 0.56306          | PCDH11Y; PCDH11X                   |
| 3    | Steroid Hormone Biosynthesis                              | 3/63    | 0.01040 | 0.60640          | SRD5A1; AKR1C4; CYP3A4             |
| 4    | Bacterial Invasion of Epithelial Cells                    | 3/78    | 0.01843 | 0.70194          | ITGB1; CLTA; SEPTIN6               |
| 5    | Cell Adhesion Molecule (CAM) Interaction                  | 4/155   | 0.02530 | 0.70194          | ITGB1; CLDN10; CADM3; NRXN1        |
| 6    | Amoebiasis                                                | 3/100   | 0.03501 | 0.70194          | CSF2; LAMB1; PLCB1                 |
| 7    | Toxoplasmosis                                             | 3/109   | 0.04342 | 0.70194          | ITGB1; IL10RA; LAMB1               |
| 8    | Endocrine and Other Factor-Regulated Calcium Reabsorption | 2/50    | 0.04975 | 0.70194          | CLTA; PLCB1                        |
| 9    | Riboflavin Metabolism                                     | 1/8     | 0.05580 | 0.70194          | RFK                                |
| 10   | FOXO Signaling Pathway                                    | 3/130   | 0.06655 | 0.70194          | SMAD3; PLK2; HOMER3                |

### 3.4 Protein–Protein Interaction (PPI) Network Analysis

Of the 143 submitted genes, 137 were mapped to STRING nodes, forming a network with 81 edges. The PPI enrichment P-value was 0.060, indicating that the network did not contain significantly more interactions than expected by chance. Connectivity was sparse: 64 of the 137 nodes (47%) had no connections, the median node degree was 1, and the maximum was 8. The most connected nodes were NRXN1 (degree 8), CSF2 and GATA3 (7 each), GFAP and CD68 (6 each), and GATA2, RGS4 and ITGB1 (5 each) (Table 4, Figure 3).

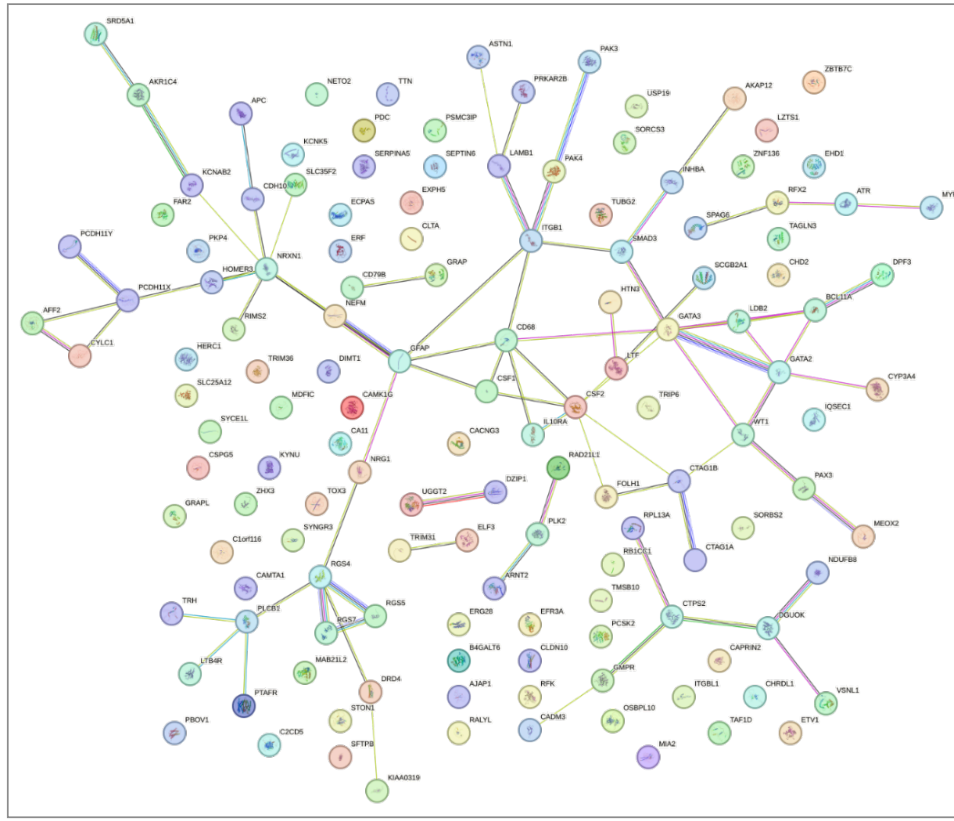


Figure 3. STRING protein–protein interaction network of the analyzed genes.

### 3.5 STRING Functional Enrichment Analysis

Functional enrichment analysis of the STRING network revealed several biological processes associated with neuronal and synaptic functions. Among the enriched functional categories were postsynapse, glutamatergic synapse, modulation of chemical synaptic transmission, and nervous system development. In contrast to the GO analysis, STRING functional enrichment, which used its default whole-genome background, reported several categories as significant after multiple-testing correction. The FDR value of the postsynapse category was about 0.00033, while the FDR value of glutamatergic synapse was around 0.0026. The FDR values of modulation of chemical synaptic transmission and nervous system development were about 0.0086. Unlike the GO and KEGG analyses, which found no significant terms, this analysis used STRING's default whole-genome background and was not repeated against a background restricted to genes on the array. (Figure 4).

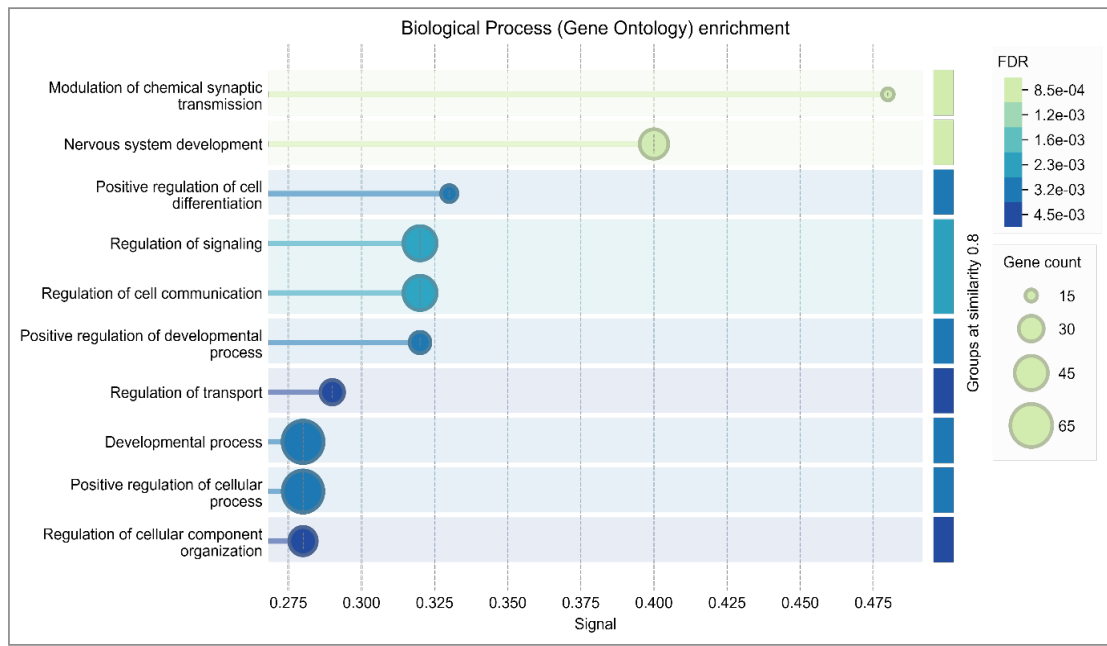


Figure 4. STRING functional enrichment analysis showing enriched biological processes associated with the analyzed gene set.

### 3.6 Network Connectivity and Candidate Hub Gene Analysis

Network connectivity analysis identified several highly connected genes within the STRING PPI network. NRXN1 showed the highest connectivity with a node degree of 8, followed by CSF2 and GATA3 with degree values of 7. GFAP and CD68 each had a degree of 6, while GATA2, RGS4, and ITGB1 each had a degree of 5. These highly connected nodes were considered candidate hub genes within the analyzed network.

Table 4. Top connected genes identified from the STRING PPI network.

| Rank | Gene    | Node degree |
|------|---------|-------------|
| 1    | NRXN1   | 8           |
| 2    | CSF2    | 7           |
| 3    | GATA3   | 7           |
| 4    | GFAP    | 6           |
| 5    | CD68    | 6           |
| 6    | GATA2   | 5           |
| 7    | RGS4    | 5           |
| 8    | ITGB1   | 5           |
| 9    | PCDH11X | 4           |
| 10   | BCL11A  | 4           |

### 3.7 Publication/Literature Enrichment Analysis

Publication enrichment analysis showed strong association of the analyzed gene set with studies related to Alzheimer's disease, hippocampal biology, and transcriptomic research. This indicates substantial overlap with previously reported literature. Because the enrichment may include studies based on GSE1297 or related analyses, it should be interpreted as literature overlap rather than independent validation. The finding therefore provides contextual support but remains exploratory.

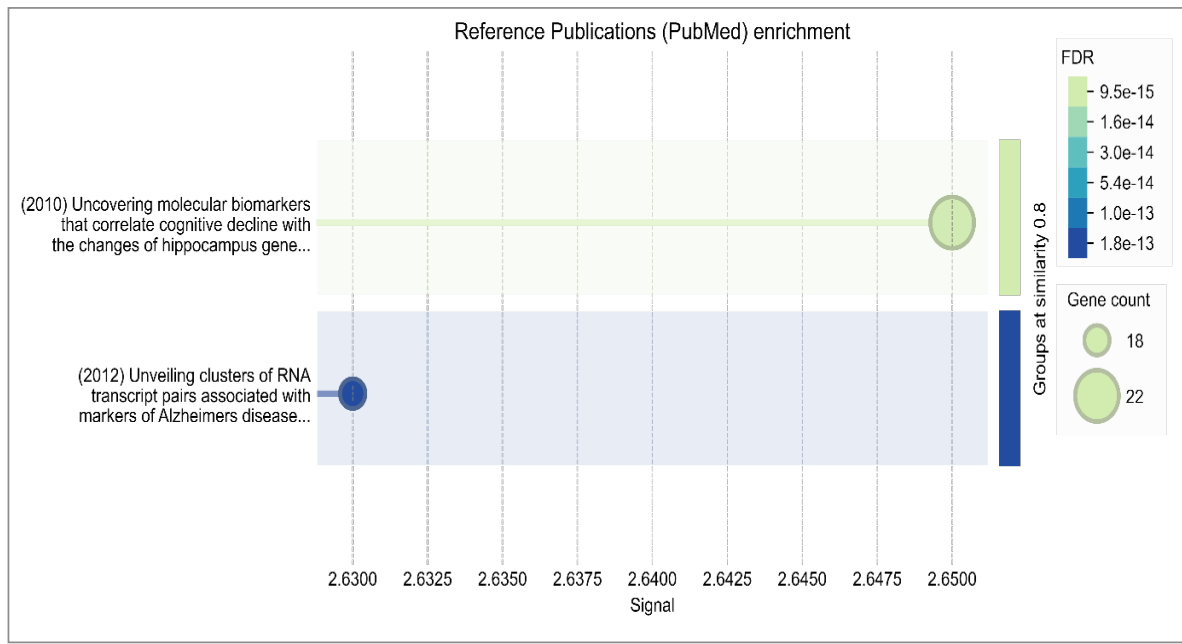


Figure 5. Publication enrichment analysis of the analyzed gene set.

## 4. Discussion

This reanalysis of GSE1297 identified no probe that differed significantly between control and severe AD hippocampus after false discovery rate correction. Although 3,482 probes had a nominal  $P < 0.05$ , the smallest adjusted  $P$ -value was 0.117. The GO and KEGG enrichment analyses were also not significant (lowest adjusted  $P$ -values 0.095 across 1,273 GO terms and 0.42 across 175 KEGG pathways), and the STRING network built from the 143 top-ranked genes was not more connected than expected by chance (PPI enrichment  $P = 0.060$ ). Only STRING's functional enrichment, which used a whole-genome background, reported significant terms, and these are interpreted cautiously below. The results are therefore exploratory and hypothesis-generating, and they do not provide statistical evidence for any individual gene or pathway.

The most likely explanation for the absence of significant genes is limited statistical power. The number of probes with a nominal  $P < 0.05$  was well above the roughly 1,114 expected by chance among 22,283 tests, which may reflect modest biological differences, unmodelled covariates such as age, sex and post-mortem interval, or a combination of these factors. A comparison of nine controls with seven severe cases cannot resolve such effects after correction for this many tests. The design may also have contributed. Blalock et al. (2004) identified their transcriptional response by correlating expression with continuous severity

measures, MMSE scores and neurofibrillary tangle counts, across all 31 subjects. Restricting the comparison to controls and severe cases discards the 15 incipient and moderate samples, and with them information about how expression changes with progression. The data therefore cannot establish which individual genes are truly differentially expressed, and the absence of significant genes should not be read as evidence that no differences exist.

The leading enrichment terms were not specifically neuronal. They concerned growth regulation, miRNA-related processes and sex-hormone biology. The top GO terms were negative regulation of cell growth, positive regulation of miRNA transcription and male gonad development, driven by genes such as SMAD3, WT1, GATA2, GATA3 and INHBA. This loosely resembles the growth-related and tumor-suppressor response described by Blalock et al. (2004), but because none of these terms was significant, the resemblance is suggestive and not confirmatory. Neuronal and synaptic terms ranked lower and rested on few genes. For example, negative regulation of dendrite development, ranked fifth, was supported by only two genes (BCL11A and KIAA0319), positive regulation of synaptic transmission ranked 26th, and nervous system development ranked 33rd (adjusted  $P = 0.18$  for both).

STRING's functional enrichment gave a different picture, reporting significant enrichment for postsynapse, glutamatergic synapse, modulation of chemical synaptic transmission and nervous system development. This result should be interpreted with care. STRING used its default whole-genome background, against which any list of genes expressed in brain tissue would be expected to look synaptic, and a custom background restricted to genes on the array was not applied, so this enrichment should be regarded as unconfirmed. The comparison for glutamatergic synapse illustrates how sensitive the result is to the method: the same category ranked 43rd and was not significant in the Enrichr KEGG analysis ( $P \approx 0.19$ ), but was significant in STRING ( $FDR = 0.0026$ ). STRING's publication enrichment showed substantial overlap with earlier studies of hippocampal expression in AD; however, because these publications may include analyses of GSE1297 itself, the finding represents literature overlap rather than independent confirmation.

Within the gene set, the most connected nodes were NRXN1, CSF2, GATA3, GFAP and CD68. NRXN1 encodes neurexin-1, a presynaptic cell-adhesion molecule that connects presynaptic and postsynaptic neurons (Südhof, 2008). GFAP is a widely used marker of astrocytes, and its up-regulation characterises reactive astrocytes (Escartin et al., 2021). CD68 is a lysosomal protein commonly used as a marker of macrophage-lineage cells, including microglia in the brain (Gottfried et al., 2008). These genes fit two well-established features of AD, synaptic loss (Selkoe, 2002) and neuroinflammation (Heneka et al., 2015), and the presence of NEFM, a neurofilament gene, echoes the neurofilament-assembly theme reported for severity-associated probes by Gómez Ravetti et al. (2010). We interpret these links cautiously for several reasons. None of these genes was individually significant. Node degree in STRING partly reflects how well studied a gene is, which favours well-characterised genes such as GFAP and CD68. The network itself was sparse: 64 of 137 nodes (47%) had no connections, and the maximum degree was 8, so the term "hub" overstates the structure. Changes in glial markers in bulk tissue can also reflect shifts in cell-type composition, such as gliosis, and not changes in expression within individual cells.

The 143-gene set also did not include the candidate hub genes reported for this dataset by the two WGCNA studies, namely the metallothioneins, NOTCH2, MSX1, ADD3 and RAB31 (Liang et al., 2018) and GRIK1 (Sun et al., 2019). This is unsurprising, since those analyses used all severity groups and a co-expression framework, whereas our list came from a nominal p-value ranking of a two-group contrast. It does show how little gene lists derived from the

same dataset can overlap when methods and cutoffs differ, and it underlines why independent validation matters before any gene is treated as a candidate.

Several limitations should be kept in mind. The sample is small, and the top-150 cutoff by nominal p-value is arbitrary and likely to include false positives. Age, sex and post-mortem variables were not modelled. The tissue was bulk, hand-dissected hippocampus, so effects cannot be attributed to specific cell types. The platform is a 2004-era microarray with limited probe coverage. Enrichr and STRING were run with their default settings and backgrounds. Post-mortem tissue also reflects late-stage disease and cannot reveal early changes. No independent replication was performed. Future work should model expression against continuous severity across all 31 samples with age and sex as covariates, apply preranked gene set enrichment analysis to the full ranking so that no p-value cutoff is needed, repeat enrichment and network analysis against a custom background of array genes, and test the findings in larger independent cohorts, ideally with single-nucleus data to resolve cell-type contributions.

## 5. Conclusion

This reanalysis of GSE1297 compared control and severe Alzheimer's disease hippocampal samples (9 versus 7) and found no individually significant differentially expressed genes after false discovery rate correction, and no significant GO or KEGG enrichment. A protein–protein interaction network built from the 143 top-ranked genes was sparsely connected, with NRXN1, CSF2, GATA3, GFAP and CD68 as the most connected nodes. These include synaptic and glial candidates consistent with known AD biology, but the findings are exploratory and are not statistically supported at this sample size. Larger, covariate-adjusted analyses with independent validation are needed to determine whether these signals are real.

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